

ABB Case Brief – Rack PDU Initiative

Case Description

ABB currently lacks an offering in the **Rack PDU (Power Distribution Unit)** market, a segment estimated at **\$2B+ globally**. Rack PDUs are socket outlet strips with protection and measurement elements used to power servers in data centers. Competitors already have strong positions, while ABB is absent in this space.

Strategic Motivation:

- Data centers are evolving toward **higher power density**, making **protection and energy management at the rack level** increasingly important.
- ABB hypothesizes that entering Rack PDUs could yield at least **\$50M/year revenue contribution within 3 years**, the minimum threshold for a strategic project.

Execution path: Three primary strategic options are under consideration — Build, Buy, or Partner. A fourth “no-go” option is also acknowledged if risks outweigh the business case.

The opportunity seems attractive, but there are key uncertainties to take into account:

- **Revenue urgency:** ABB must start capturing significant revenues in 2–3 years, given market maturity.
- **Technology velocity:** Some internal voices suggest technology could change so drastically in the next 2–3 years that there may not be rack PDUs anymore. Migration to higher rack loads (from 10–20kW to 100–200kW) or DC distribution in data centers could disrupt the market.
- **Channel mismatch:** Rack PDUs are sold largely through IT distribution channels (server OEMs, integrators), not ABB’s traditional electrical sales force. ABB lacks a dedicated sales organization in this space.
- **Acquisition complexity:** ABB highlighted risks of acquiring small, fragmented players in remote geographies, which could bring organizational or integration challenges. Experts also suggest ABB should critically assess its **track record in M&A (and post-merger integration or lack thereof)**, especially within Smart Buildings.

- **Market accessibility:** The **headline market size may be misleading**. For example, while China's rack PDU market is valued at ~\$500M, the **accessible market for foreign companies is only ~\$75M** due to restrictions and barriers. Similar constraints exist in India and other geographies.
- The opportunity must also be judged in terms of **strategic criticality** — i.e., why it matters in ABB's broader innovation and growth story, especially compared against other options in the portfolio pipeline.

Decision responsibility sits with the **Energy Distribution Business Line** (Javier Gomez and Stefano Besana), with oversight from the **Division CFO and President**.

Scope of Analysis (per ABB's framing document):

- **Portfolio & roadmap assessment:** How does Rack PDU fit with ABB's broader energy distribution portfolio? What internal capabilities or synergies can ABB leverage?
- **Market & segment evaluation:** Target segments (enterprise, hyperscale, colocation), geography breakdown, growth trajectories, and **accessible market size by geography**.
- **Go-to-market pathways:** Direct vs partner channels, IT vs electrical distribution sales routes.
- **Scenario evaluation:** Acquisition, Internal Development, Partnership/Brand Label.
- **Outcome expectation:** Deliver a clear view on what are the pros and cons for entering the market and what is the optimal scenario for market entry (best of Build/Buy/Partner)

Decision Framing Document – ABB Solid-State Circuit Breaker Initiative

1. Context

ABB has developed a **solid-state circuit breaker (SSCB)** product. However, the **market demand and adoption curve remain unclear**. The strategic challenge is whether ABB should actively pursue this as a future growth area within its **Electrification and more specifically ABB Smart Buildings division**.

2. Decision Statement

Core Question:

“Should ABB pursue solid-state circuit breakers as a strategic business, and is the current offer in the Power Division the correct approach?”

3. Options (MECE)

- **Option A – Go Big:** Scale investment in SSCB, commit to commercialization, and build GTM strategy.
 - **Option B – Explore Selectively:** Continue R&D and pilot projects while monitoring market signals.
 - **Option C – Partner/Wait-and-See:** Collaborate with ecosystem players to shape the market; limit direct exposure.
 - **Option D – No-Go:** Exit SSCB, redeploy resources to more proven opportunities.
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4. Objectives

1. Establish whether SSCB is a **strategically critical technology** for ABB’s long-term positioning.
 2. Identify potential market or technological trends that may drive SSCB adoption in certain specific market segments and identify segments that have the biggest promise for adoption
 3. Are there opportunities to Achieve **positive ROI** within a defined horizon (e.g., 5–7 years).
 4. Will this technology open the market for new players that may disrupt the current incumbents
 5. Ensure ABB’s **Smart Building Division offering is competitive and aligned** with future demand.
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5. Evaluation Criteria

- **Market Adoption Potential** (segment size, readiness) – weight 30%
 - **Strategic Fit** (alignment with ABB’s Power Division portfolio) – weight 25%
 - **Technology Maturity** (reliability, scalability, cost curve) – weight 20%
 - **Time-to-Market / Competitive Position** – weight 15%
 - **Financial Return Potential** – weight 10%
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6. Assumptions

- SSCB technology offers **superior performance** (speed, safety, efficiency) compared to mechanical breakers.
 - Market adoption is **uncertain**; no established large-scale demand yet.
 - ABB has **internal R&D capability** but limited clarity on commercialization path.
 - Competing firms may also be exploring SSCB but no dominant leader exists.
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7. Dependencies & Interactions

- Adoption depends on **downstream applications** (main focus - datacenter, alternative and parallel pathway is in renewables).
 - ABB’s positioning depends on **whether SSCB is integrated into broader ABB power products**.
 - Early pilot customers could shape technology roadmap and standards.
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8. Risks

- **No Market Emerges:** Likelihood: Medium | Impact: High.
 - **High R&D Cost with Delayed ROI:** Likelihood: Low | Impact: Medium.
 - **Misalignment with ABB Portfolio:** Likelihood: Low Impact: High.
 - **Competitors Capture Early Standardization:** Likelihood: Medium | Impact: High.
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9. Constraints

- Limited clarity on **market readiness and timing**.
- Must align within ABB Power Division’s budget and strategy.

- ABB cannot sustain indefinite R&D without a clear commercialization path.
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10. Information Gaps

- **Market Segmentation:** Which industries would be early adopters of SSCB?
- **Willingness to Pay:** What premium (if any) are customers willing to accept vs mechanical breakers?
- **Technology Roadmap:** What is the maturity curve and cost trajectory for SSCB? Which “necessary and sufficient” features would tip customers towards a solid state solution?

Where the SSCB semiconductor can potentially make the difference:

- Fault level during short circuit will become irrelevant while today is a big limitation factor for the standard technology
 - Speed for short circuit clearance will be in the range of few us:
 - Safe for the system: fault cleared in no time with no solicitation on distribution system and devices
 - Safe for users: in case of accidental contact voltage is removed in no time
 - Easier scalability with voltage increase. Working beyond 500-750VDC is still a great challenge for traditional technology, brilliantly overcome by semiconductors
 - Combined protection and actuator (potentially needed for load shedding or automations) with almost infinite load operations available and no mechanical elements moving
 - Capability to “pre energize “ the network smoothly after full shut-down
 - Capability to stabilize the fault during short circuit to a level where other parts of the circuit should remove the fault
 - **Competitive Landscape:** Who else is developing SSCBs, and at what stage?
 - **Regulatory Standards:** Are there standards emerging that could accelerate or slow adoption?
 - **Risk appetite:** What is ABB willing to bet to become the leading player in this field \$1-2,5 million
 - **System perspective advantages and risks:** how would a solid state breaker fit from a system perspective? Is it a straight replacement for an electromechanical breaker or are there potential side effects from SSC operation?
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11. Timeline & Scope

- Simulation horizon: 5–7 years (longer adoption curve).
 - Decision point: Define **go/no-go within next 12 months** based on market validation.
 - Scope: Limited to SSCB (not broader breaker portfolio).
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12. Stakeholders

- **Decision Owner:** ABB Power Division President.
 - **Input Providers:** R&D leadership, Business Line Leaders.
 - **External Stakeholders:** Key pilot customers, regulators, ecosystem partners.
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13. Decision Type

- **Strategic.**
 - Medium irreversibility (investment sunk costs, positioning).
 - Stakes: **Moderate-to-high**, given potential disruptive impact if SSCBs become mainstream.
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Key Stakeholder Questions:

Technology & Commercialization

- How mature is ABB's current SSCB prototype/product compared to competitors? Has any research has been done here?
- Should we be aware of any **technical or cost barriers** preventing adoption?
- Would ABB plan to **commercialize directly** or pursue **ecosystem/partnership plays** to de-risk adoption or is it unclear/undecided at the moment?

Investment & Risk Appetite

- What **ROI timeline** is ABB willing to tolerate (5–7 years vs shorter)?
- Is ABB open to pursuing SSCB as a **strategic bet with smaller immediate payback**, or does it need clearer near-term economics?
- What level of **capital allocation** (R&D, pilots, partnerships) is acceptable before a go/no-go decision?

Hypotheses to Test

1. **SSCBs will become mainstream in critical segments** (e.g., datacenters, EV fast charging, renewables) within 5–7 years.
2. **Early adoption will depend on total cost of ownership (TCO)** rather than upfront price, meaning ABB must position SSCB as a **performance + lifecycle cost** play.
3. ABB's SSCB offering is **best pursued in conjunction with existing ABB products** (power distribution, monitoring, digital layers), not as a standalone breaker.
4. ABB cannot afford a long internal development cycle without **external partners/pilot customers** driving credibility.
5. The risk of **missing the adoption curve** (if SSCBs take off) outweighs the risk of **investing too early** with measured exposure.

Information Gaps to Close Before Proceeding

- **Market Segments:** Which verticals are willing to be first movers?
- **Customer Economics:** How much premium would customers accept over mechanical breakers?
- **Competitive Landscape:** Who else (GE, Schneider, startups) is developing SSCBs, and where are they on readiness? Are there other technologies on the horizon?
- **Cost Curve:** How fast can SSCB costs drop with scaling/innovation?
- **Standards & Regulation:** Are there pending standards or mandates that could accelerate adoption?

- **Integration Path:** How does SSCB fit into ABB's existing Power Division sales channels and solutions?

